

GIOVANNI TAMMARO

AWS GenAI Engineer | AI Infrastructure

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Portfolio & GitHub: <https://giovannitammaroaws.github.io/portfolio/>

Medium: <https://medium.com/@giovannitammaro76>

Professional Summary

AWS-Certified Cloud & AI Engineer with 20+ years of infrastructure experience, including 6 years at Amazon Web Services (Dublin). From January 2022 to August 2026, I have focused on building production-oriented GenAI systems on Amazon Bedrock, including RAG pipelines, vector retrieval, LLM routing, grounded responses, evaluation patterns, security guardrails, and cost-aware operations. Strong in Python, FastAPI, AWS-native architecture, and end-to-end delivery from design to production.

Skills

Cloud & AI: AWS (Bedrock, Lambda, S3, API Gateway, EC2, IAM, Route 53, CloudFront)

GenAI & Retrieval: LLMs, RAG, Bedrock Knowledge Bases, pgvector, Prompt Engineering, AI Agents, Evaluation

Backend & Delivery: Python, FastAPI, PostgreSQL, DynamoDB, SQS, GitHub Actions, CI/CD, Docker, CloudWatch, X-Ray, Git

Security & Governance: IAM, KMS, WAF, GuardDuty, Cognito, Inspector, DevSecOps

Work Experience

AWS AI & AI Infrastructure Engineer - Independent Projects

Freelance | Remote

Jan 2022 - Present

- Built and evolved an AWS-hosted RAG platform with a React frontend on CloudFront and a containerized FastAPI backend on ECS Fargate, connected to Aurora for end-to-end ingestion workflows, LLM integrations, and operational visibility.
- Expanded ingestion to support both text-based and scanned PDFs, adding Docling and AWS Nova Pro-based document understanding to improve semantic extraction and retrieval quality on low-coverage documents.
- Revamped the ingestion layer with automated reconciliation, gap analysis, and real-time monitoring, improving data freshness, pipeline visibility, and downstream AI reliability.
- Strengthened observability with AWS CloudWatch and X-Ray, ingesting telemetry and traces for real-time ingestion monitoring, threshold-based alerting, and faster issue detection across the RAG pipeline.
- Improved RAG latency and retrieval efficiency through caching and retrieval optimizations, including pg_prewarm to fully warm the HNSW index, delivering about 80% faster semantic-search response times on retrieval paths.
- Designed metadata editing, audit-trail, and collection-scoping capabilities, improving transparency and enabling more precise queries across targeted financial datasets.
- Introduced RAG evaluation workflows with RAGAS to measure retrieval and answer quality, support regression detection, and improve overall RAG reliability.

Network Automation Engineer

Fiserv | Ireland

Aug 2021 - Dec 2021

- Refactored Django/JS stack at Fiserv, cutting page load times by 40-60%, PostgreSQL CPU usage by ~30%, and API response times to under 200ms.
- Automated network infrastructure deployments with Ansible playbooks, cutting per-site rollout time from weeks to hours and eliminating manual configuration errors.

Network Development Engineer II

Amazon Web Services (AWS) | Dublin, Ireland

Apr 2017 - Aug 2021

- Engineered internal self-service tooling (React + Python) adopted by Amazon's global datacenter network teams - eliminating hours of manual operations per week per deployment cycle.
- Designed and delivered a zero-touch pre-provisioning platform (Python, EC2, S3) for L7 hardware - reducing per-rack deployment time from weeks to hours across thousands of devices.
- Optimized internal dashboards with multithreaded ingestion logic, increasing throughput by ~40%.

Network Deployment Engineer I

Amazon Web Services (AWS) | Dublin, Ireland

Aug 2015 - Apr 2017

- Scaled Amazon's global datacenter network under the Aggregation Scaling & Deployment program, increasing capacity by ~30% via spine scaling templates across multiple regions.
- Developed Python automation scripts and Jinja templates for network provisioning workflows - reducing pre-deployment manual steps and accelerating network readiness across global datacenters.
- Resolved bugs in global network automation systems with fleet-wide impact; contributed unit-tested improvements to shared internal Python libraries.

Network & Systems Engineer

Parker Hannifin | Milan, Italy

Apr 2004 - Jul 2015

- Led EU/UK datacenter consolidation and AD migrations, cutting authentication latency by 40%.
- Reduced workstation provisioning time from 40 min to under 5 min via VMware automation.
- Automated Cisco provisioning and L2 security hardening across EU/UK sites: DHCP Snooping and Spanning Tree.

Security Consultant

Freelance | Milan, Italy

Jan 2002 - Apr 2004

- Implemented and tuned enterprise security stack (IDS/IPS, endpoint protection, incident response) across on-premises networks for SMB clients.
- Advised clients on L2 network segmentation design and switch hardening - delivering new VLAN architectures, access control policies, and L2 security configurations across on-premises environments.

Projects (All on GitHub)

- Atlas AI RAG Platform - Financial Document Q&A on AWS
- Built a production-oriented GenAI platform on Amazon Bedrock with pgvector retrieval, grounded answers, provider routing, and cost-aware inference.
- Added event-driven ingestion, observability, and security controls for an end-to-end AWS AI use case.

Certifications

AWS Certified Generative AI Developer - Professional

AWS Certified Solutions Architect - Associate

AWS Certified AI Practitioner

AWS Knowledge Badge: Amazon EKS

Cisco Certified Network Professional (CCNP)

Languages

Languages: Italian (Native), English (Fluent), Spanish (Basic), French (Basic), Portuguese (Basic)